INT255:MATHEMATICS BEHIND MACHINE LEARNING

L:2 T:0 P:2 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: explain vector spaces, norms, and linear operators used in machine learning models.

CO2 :: apply matrix decompositions to perform dimensionality reduction and representation learning.

CO3 :: analyze probability-based loss functions and estimation techniques used in machine learning.

CO4 :: apply optimization techniques to train machine learning models efficiently.

CO5 :: analyze margin-based classifiers using mathematical optimization principles.

CO6 :: evaluate regularization and generalization trade-offs in machine learning models.

Unit I

Linear Algebra Foundations for Machine Learning : Vectors, matrices, and tensors in machine learning, Vector spaces and subspaces, Norms (L1, L2) and projections, Linear operators and transformations in ML

Unit II

Matrix Decompositions for Representation Learning : Eigen decomposition and its limitations in ML, Singular value decomposition (SVD), Principal component analysis (PCA) from a geometric and optimization perspective, Linear discriminant analysis (LDA), Applications of matrix factorization in recommendation systems

Unit III

Probability Theory and Loss Functions for Machine Learning : Random variables in ML models, Probability distributions used in learning algorithms, Likelihood estimation in ML context, Loss functions: squared error, logistic loss, Bayesian interpretation of learning models

Unit IV

Optimization Techniques for Machine Learning : Optimization problem formulation in ML, Convex sets and convex functions, gradients and directional derivatives, Gradient descent and variants (batch, stochastic, mini-batch), Momentum-based optimization, RMSProp and Adam, Optimization considerations in large-scale ML systems

Unit V

Mathematical Foundations of Support Vector Machines : Geometric interpretation of classification margins, hard margin and soft margin SVM, Primal and dual optimization problems, Lagrangian formulation, Kernel trick and kernel functions, Optimization perspective of SVM training

Unit VI

Regularization and Generalization Theory : Bias-variance trade-off, Overfitting and underfitting from a mathematical viewpoint, L1 and L2 regularization, Norm-based constraints, Structural risk minimization, VC dimension (intuition)

List of Practicals / Experiments:

List of Practicals

- 1.Represent a machine learning dataset using vectors and matrices and perform basic linear algebra operations.
- 2.Compute L1, L2, inner products, and vector projections to analyze feature similarity.
- 3.Apply linear transformations to data and study their effect on feature embeddings.
- 4.Perform Singular Value Decomposition (SVD) and construct a low-rank approximation of a data matrix.
- 5.Implement Principal Component Analysis (PCA) using covariance matrices for dimensionality reduction.
- 6.Model data using probability distributions and evaluate likelihood and common loss functions.
- 7.Formulate and solve a machine learning optimization problem using gradient descent methods.

- 8.Analyze convergence behavior of momentum-based and adaptive optimization algorithms.
- 9.Implement Support Vector Machine classification and study margin maximization using kernel functions.
- 10.Demonstrate overfitting, underfitting, and the effect of L1 and L2 regularization on model generalization.

Text Books:

1. MATHEMATICS FOR MACHINE LEARNING by MARC PETER DEISENROTH, A. ALDO FAISAL, AND CHENG SOON ONG, CAMBRIDGE UNIVERSITY PRESS

References:

1. INTRODUCTION TO LINEAR ALGEBRA by GILBERT STRANG, Wellesley-Cambridge Press